MIL-P-48349(PA)
26 August 1974

MILITARY SPECIFICATION

PROPELLANT M30A1
FOR USE IN 155MM PROPELLING CHARGES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers M30A1 Propellant for use in Cartridges employed with the XM198 Cannon prior to DT/OT II Testing.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-652 - Propellants Solid for Cannons Requirements and Packing
MIL-STD-1168- Lot Numbering of Ammunition
MIL-STD-1171- Acceptance and Description Sheets for Propellants and Explosives
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective

FSC: 1376

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:22Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

Title 49, Department of Transportation: Parts 0-199

AMC PAMPHLET 706-114

Experimental Statistics, Section 5, Tables

The Code of Federal Regulations is available from the Superintendent of Documents, Government Printing Office, Washington,D.C., 20402. Orders for the above publication should cite: 49 CFR 0-199 (latest revision).

3. REQUIREMENTS

3.1 Materials. The propellant shall be M30A1, Type II conforming to the requirements of MIL-STD-652, except para 4.2.3.3.1.2.

3.2 Proving Ground. The propellant, when fired in the XM198 Cannon and with the weight of projectile (96 pounds (lbs) for which it was manufactured, shall comply with the following requirements when tested as specified in 4.3.

3.2.1 Assessment. At 70 degrees Fahrenheit (°F) and standard weapon conditions, the propellant shall be capable of charge assessment to a velocity of 2710 feet per second (fps) without an associated expected pressure greater than 47500 pounds per square inch (psi).

3.2.2 Uniformity of velocity. The standard deviation in velocity for the lot at 70°F shall not exceed 6.5 fps.

3.2.3 Permissible Individual Maximum Pressure. The propellant at recommended charge weight shall not produce a chamber pressure greater than 58,500 psi at any operating temperature (-50°F to +145°F) when fired in a standard weapon.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:22Z
Check the source to verify that this is the current version before use.

3.3 Straightness. Twenty seven pounds of stick propellant, cut to the length specified in the contract, shall fit into a 6.1 inch inside diameter cylinder. In addition, each of these individual sticks of propellant shall not bow more than .25 inch.

3.4 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. No reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification unless otherwise specified.

3.5 Physical Properties. The following physical properties shall be determined as specified in 4.3.6 and reported for information (see 6.3):

a. Density
b. Compressive strength at 77 $\pm$ 4°F
c. Deformation at maximum load.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109, in order to define terms used herein. The provisions of MIL-A-2550 shall apply,

4.1.1 Submission of product. At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:22Z
Check the source to verify that this is the current version before use.

b. Number of batches of product inspected (shall be made available upon request by the contracting officer).

c. Results obtained, by defect code, for all inspections performed (shall be made available upon request by the contracting officer)

d. Drawing, specification number, and date, together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contracting documents (shall be made available upon request by the contracting officer).

f. Number of items in the lot.

g. Date submitted.

h. Eight copies of propellant description sheet SMU Form 1047 containing the applicable data (see 6.6)

4.1.2 The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation. The term "lot" as used throughout this specification, refers to an inspection lot, which is defined as an essentially homogeneous quantity of propellant from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. A lot shall consist of one or more sub-lots or one or more batches of M30A1 propellant produced by one manufacturer, in accordance with the same specification, and same specification revision, under one continuous set of operating conditions. ~~Each lot shall consist~~ of that quantity of M30A1 propellant which has been subjected tothe same unit chemical or physical process intended to make the final product homogeneous.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:22Z
Check the source to verify that this is the current version before use.

4.2.2 Testing

4.2.2.1 Materials (see 3.1). Each lot of propellant shall be sampled and tested for composition, form, dimensions and chemical-physical properties in accordance with MIL-STD-652.

4.2.2.2 Proving Ground (see 3.2). Each lot of propellant shall be sampled for proving ground tests in accordance with MIL-STD-652.

4.2.2.2.1 Assessment (see 3.2.1) Informational. The lot shall be referred if the assessed charge weight of propellant has an expected pressure greater than 47,500 psi (see 6.8)

4.2.2.2.2 Velocity Uniformity (70°F) (See 3.2.2) Informational. If the product of the factor (see 6.4) times the velocity standard deviation of the rounds tested in accordance with 4.3.3 exceeds the applicable requirement the lot shall be referred (see 6.8).

4.2.2.2.3 Individual pressure (see 3.2.3) Informational. If any individual pressure observed during uniformity firings at any operating temperature (-50°F to 145°F) after correction for difference between as fired and recommended service charge weight exceeds 58,500 psi, the lot shall be referred (see 6.8).

4.2.2.3 Straightness (see 3.3). Ten samples each containing 27 pounds of stick propellant, shall be randomly selected from each lot of propellant and tested for straightness. If any of the ten samples, or individual sticks, fails the straightness requirement, an additional ten samples shall be selected and tested for straightness. If any of these samples fail the straightness requirement, the lot shall be rejected.

4.3 Test methods and procedures

4.3.1 Proving ground tests. These tests shall be conducted at a Government proving ground (designated by the contracting officer) in accordance with the applicable Proving Ground Acceptance Test Procedure.

4.3.2 Charge assessment

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:22Z
Check the source to verify that this is the current version before use.

4.3.2.1 Charge Slope Method. Three or more groups of rounds loaded with selected charge weights of propellant (weights selected should bracket expected service charge weight) shall be conditioned and fired at 70°F. From the data obtained, a charge weight shall be calculated which will yield the prescribed service velocity. This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.3.3). The difference between the corrected velocity obtained during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the velocity charge weight curve previously

4.3.2.2 Constant Slope Method. After ten (10) lots of a given production series have been fired to provide a good knowledge of the propellant characteristics, the charge slope method of assessment may be discontinued and the charge weight determined by the constant slope method. A composite velocity versus charge weight versus pressure curve can be established from test results generated. This curve can then be used as in 4.3.2.1 above to calculate the charge weight needed to achieve service velocity. This calculated charge weight is then to be loaded into rounds for the uniformity series and final adjustment is to be made as indicated above for the charge slope method.

4.3.3 Uniformity series

4.3.3.1 145°F. Thirty rounds using Stick Propellant Charges shall be assembled with the assessed charge of propellant, conditioned to 145°F (as indicated by a thermocoupled control round (rnd)) and fired alternately with calibration rounds. The velocity and pressure means as well as the standard deviations associated with these parameters shall be calculated. All results shall be corrected to recommended charge weight conditions (i.e. difference between as fired charge weight and recommended charge weight).

4.3.3.2 70°F. Thirty rounds using Stick Propellant Charges shall be assembled with the assessed charge of propellant, conditioned to 70°F (as indicated by thermocoupled control rnd) and fired alternately with calibration rounds. The velocity

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:22Z
Check the source to verify that this is the current version before use.

and pressure means, as well as the standard deviations associated with these parameters shall be calculated. All results shall be corrected to standard shell weight (96 lbs) and standard weapon/firing conditions (i.e. the difference between calibration performance under test conditions and standard conditions) and to recommended charge weight conditions.

4.3.3.3 -50°F. Thirty rounds using Stick Propellant Charges shall be assembled with the assessed charge of propellant, conditioned to -50°F (as indicated by a thermocoupled control rnd) and fired alternately with calibration rounds. The velocity and pressure means as well as the standard deviations associated with these parameters shall be calculated. All results shall be corrected to recommended charge weight conditions (i.e. difference between as fired charge weight and recommended charge weight).

4.3.3.4 Pressure/Time Evaluation (for information only).Ten Stick Propellant Charges each shall be fired at -50°F, +70°F and +145°F for pressure versus time evaluation in an instrumented XM198 Cannon where Piezo pressure versus time and ignition delay is recorded in the breech spindle and at a point 45.5 inch RFT (tapped in cannon tube). Muzzle velocity and pressures shall also be recorded.

4.3.4 Refiring. If for any reason the proving ground considers that test conditions have detrimentally affected the test result, additional charges as required shall be tested.

4.3.5 Straightness. Test bundles of stick propellant shall be placed into a 6.1 inch ID rigid cylinder having a length of 25 inches minimum without the use of tools such as hammers or wedges. Stick straightness shall be measured by rolling the stick on a flat plate and measuring the maximum distance observed between the plate and the stick.

4.3.6 Physical Properties (Informational)

4.3.6.1 Density. The propellant specific gravity test shall be conducted on each lot in accordance with Method 510.1 of MIL-STD-286, except that a 10 ml pycnometer may be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:22Z
Check the source to verify that this is the current version before use.

4.3.6.2 Compressive strength. Prepare one test specimen from each of five propellant sticks per lot. Each specimen shall be machined to a length of 0.375 ± .005 inch with specimen ends perpendicular to the specimen outside diameter within 0.005 inch. A compression test shall be conducted on each specimen at 77 ± 4°F. The compressive load shall be applied at a maximum rate of 1 inch/inch gage length/minute, until the specimen fails. Failure shall be that point at which the load is maximum. The original cross sectional area shall be determined for each specimen prior to test. The compressive strength shall be determined for each specimen as follows:

$$\text{Compressive stress} = \frac{\text{Applied load at failure}}{\text{Original cross sectional area}}$$

4.3.6.3 Deformation. The deformation at the point of maximum load shall be determined on the specimens tested in 4.3.6.2, and shall be expressed as:

$$\% \text{ Deformation} = \frac{(\text{Original length} - \text{length at max. load}) \times 100}{\text{Original length}}$$

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. Packing and marking shall be in accordance with MIL-STD-652 or other instructions provided by the contracting officer.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following.

6.1.1 Procurement Requirements

a. Title, number and date of this specification.
b. Packing required (see 5.1)
c. Process. Details of the manufacturing process and the equipment used by the Contractor will be submitted to the procuring activity in writing prior to commencement of manufacture.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:22Z
Check the source to verify that this is the current version before use.

6.1.2 Contract Data Requirements. When required by incorporation of DD Form 1423 into the contract, Acceptance and Description sheets shall be prepared for each lot in accordance with MIL-STD-1171.

6.2 Standard deviation. The standard deviation shall be calculated with (n-1) as the devisor in a standard statistical technique equivalent to that shown in MIL-STD-414, Section B, Example B-1.

6.3 Submission of test data. In addition to the normal distribution of records, one copy of all informational test data shall be forwarded to: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-P, and one copy to ATTN: SARPA-ADEP-3, Dover, New Jersey 07801.

6.4 Factors. The factors, as taken from Table A-21 of AMC Pamphlet 706-114 using the 95 percent upper confidence limit (A.05) and n-1 degrees of freedom, make allowances for the probability that a standard deviation from a smaller sample size exceeds the true standard deviation by chance alone.

6.5 Tolerance. A tolerance of -5°F applies to all -50°F and 145°F requirements and a tolerance of plus or minus 2°F to all 70°F requirements.

6.6 Submission of SMU Form 1047. Information contained in propellant description sheet, SMU Form 1047, need not be resubmitted in the form of a certificate provided the provisions regarding certifying of responsible agents is adhered to (see 4.1.2).

6.7 Surveillance sample. Samples shall be forwarded to Picatinny Arsenal, Dover, N.J. ATTN: SARPA-FR-G, for the propellant surveillance test program in accordance with MIL-STD-652.

6.8 Referred lots. Propellant lots shall be referred to Picatinny Arsenal, Dover, N.J. 07801, ATTN: SARPA-ADEP-3 and SARPA-QA-A-P.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1376-A025

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:22Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:22Z
Check the source to verify that this is the current version before use.